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EVLKST8500GH-2 G3-Hybrid PLC&RF connectivity development kit User Terminal quick start guide

October 2023

Introduction

- Scope of this presentation is to introduce the user Terminal available on the USB port of EVLKST8500GH-2 (abbreviated, EVLKST8500) STM32 Nucleo board.
- This terminal allows user to get into the evaluation of some features like UDP data transfers via some launchable tests/demo.

- Required elements:
  - PC with a terminal tool installed (e.g.Tera Term).
  - Get 2 of EVLKST8500 boards to run data transfers from one PLC/RF node to the other.
  - Update boards with latest STSW-ST8500GH-2 package available at: https://www.st.com/content/st_com/en/products/embedded-software/evaluation-tool-software/stsw-st8500gh-2.html

Sections

1 Preliminary steps

2 General handling

3 UDP tests

4 UDP connections

5 Connected devices

6 UDP image transfer

7 Manage SPI Flash memory

8 Kick/Connect/Disconnect

9 GMK update

10 Fast Restore

11 Reset

12 Last Gasp

13 MAC test

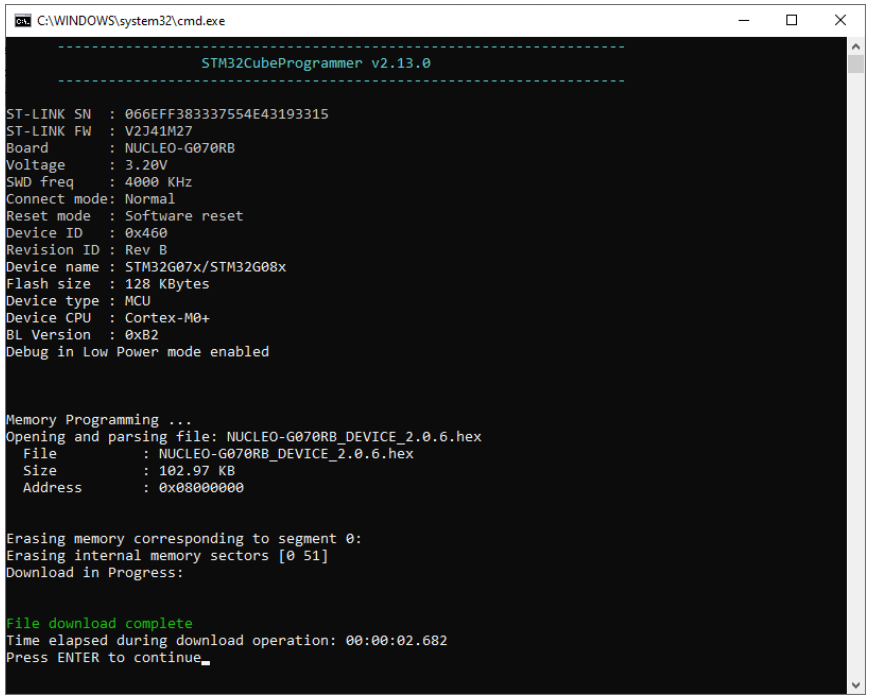
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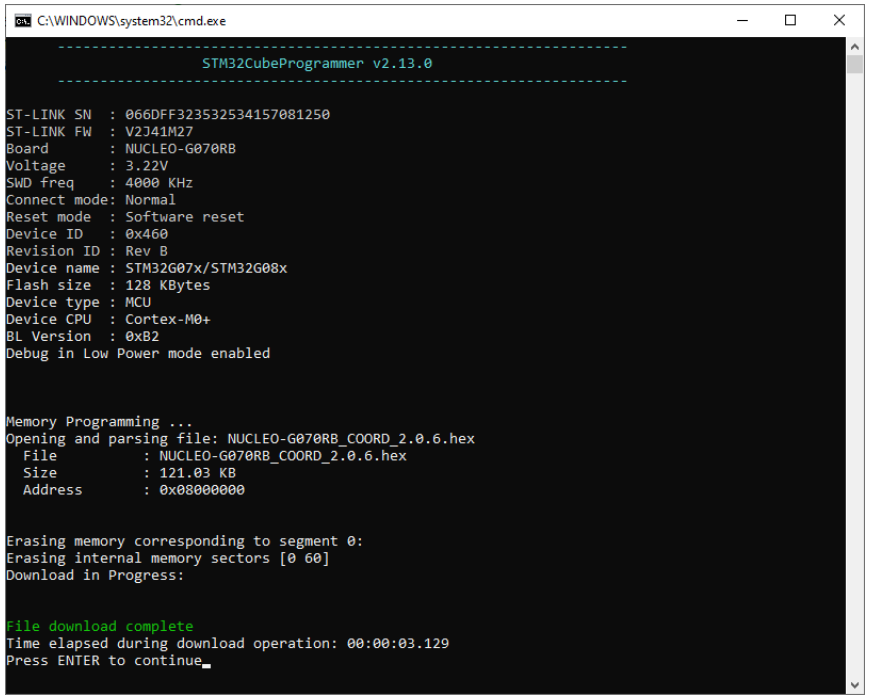
Preliminary steps

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Nucleo board FW upgrade

- After installing STM32CubeProgrammer, connect the USB cable to the Nucleo board USB connector and launch the appropriate batch (.bat) file, based on the Nucleo model and the network role (NUCLEO-G070RB, NUCLEO-G474RE or NUCLEO-L476RG; Coordinator or Device).
  - Please note that for each G3-Hybrid network exactly one node must be programmed as Coordinator.
  - Please connect and program one EVLKST8500GH-2 at a time.

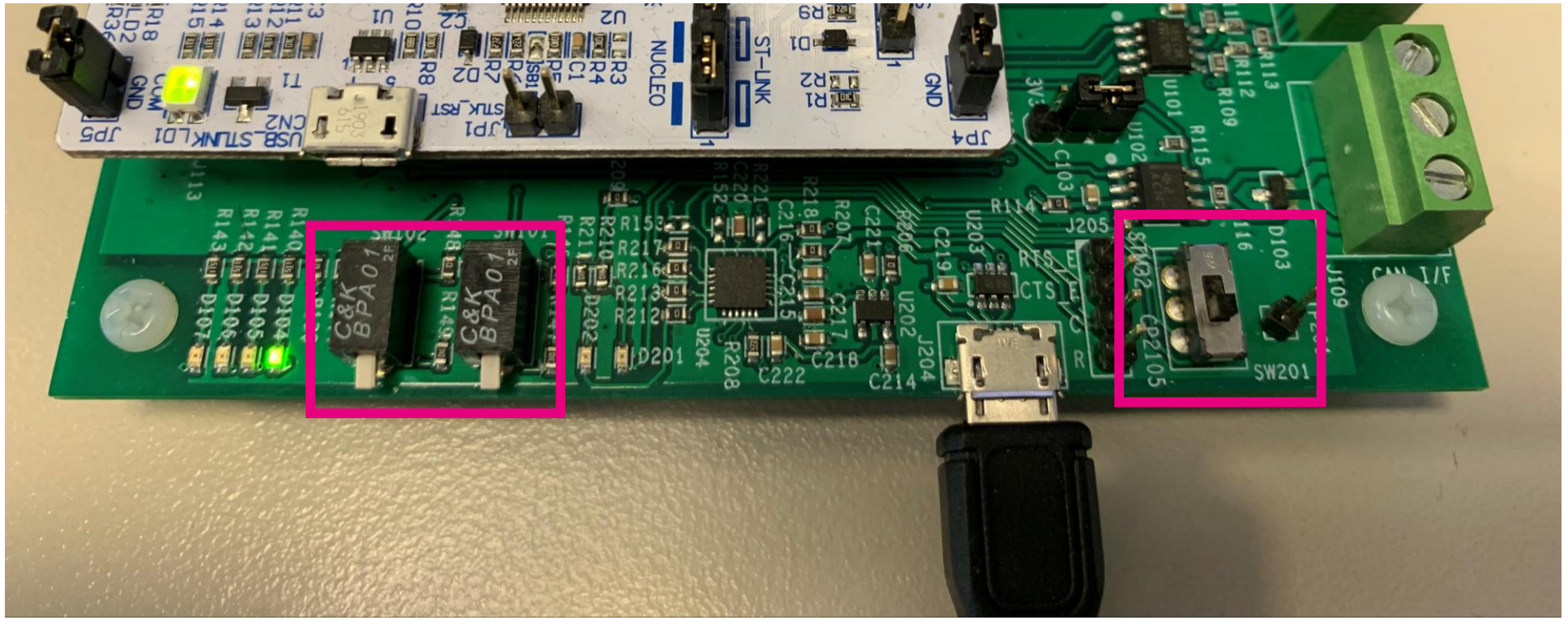




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ST8500 FW upgrade preparation

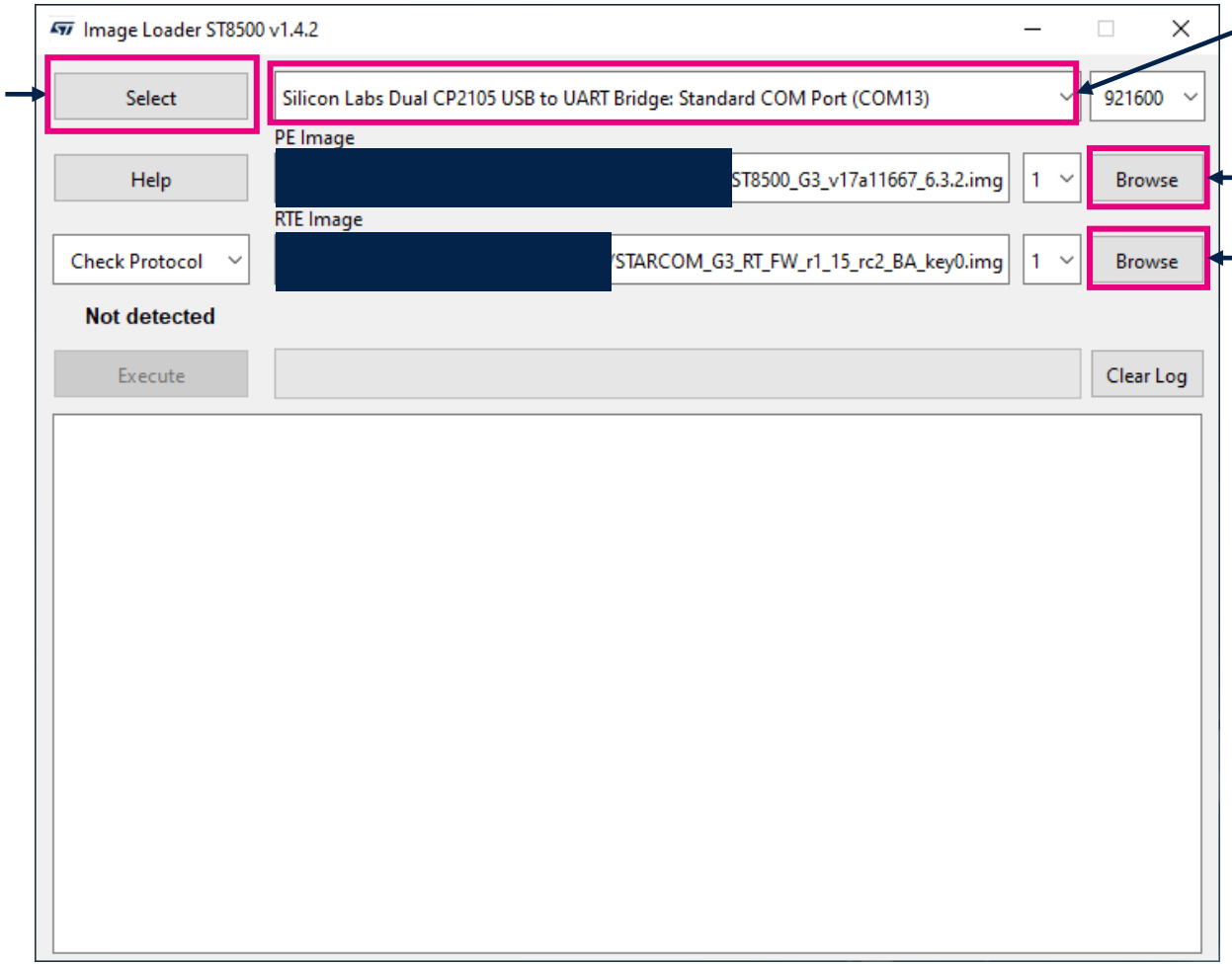
- Connect the PC to the USB socket on the **motherboard**, set SW101 low and SW201 to **CP2105**



ST8500 FW upgrade execution 1/3

- Launch Image Loader GUI (*start_gui.vbs*, inside *ST8500_G3_v17a11667_6.3.2\G3_Images_v17a11667_6.3.2\image_loader_st8500_v1.4.2.0\dist_python*) and follow the illustrated steps*.

4. Click on “Select”



1. Select the COM port (Standard)

2. Select the PE image

3. Select the RTE image

*It is necessary to install Python 3.11 (or more recent) x64 bit and the modules listed in *requirements.txt*

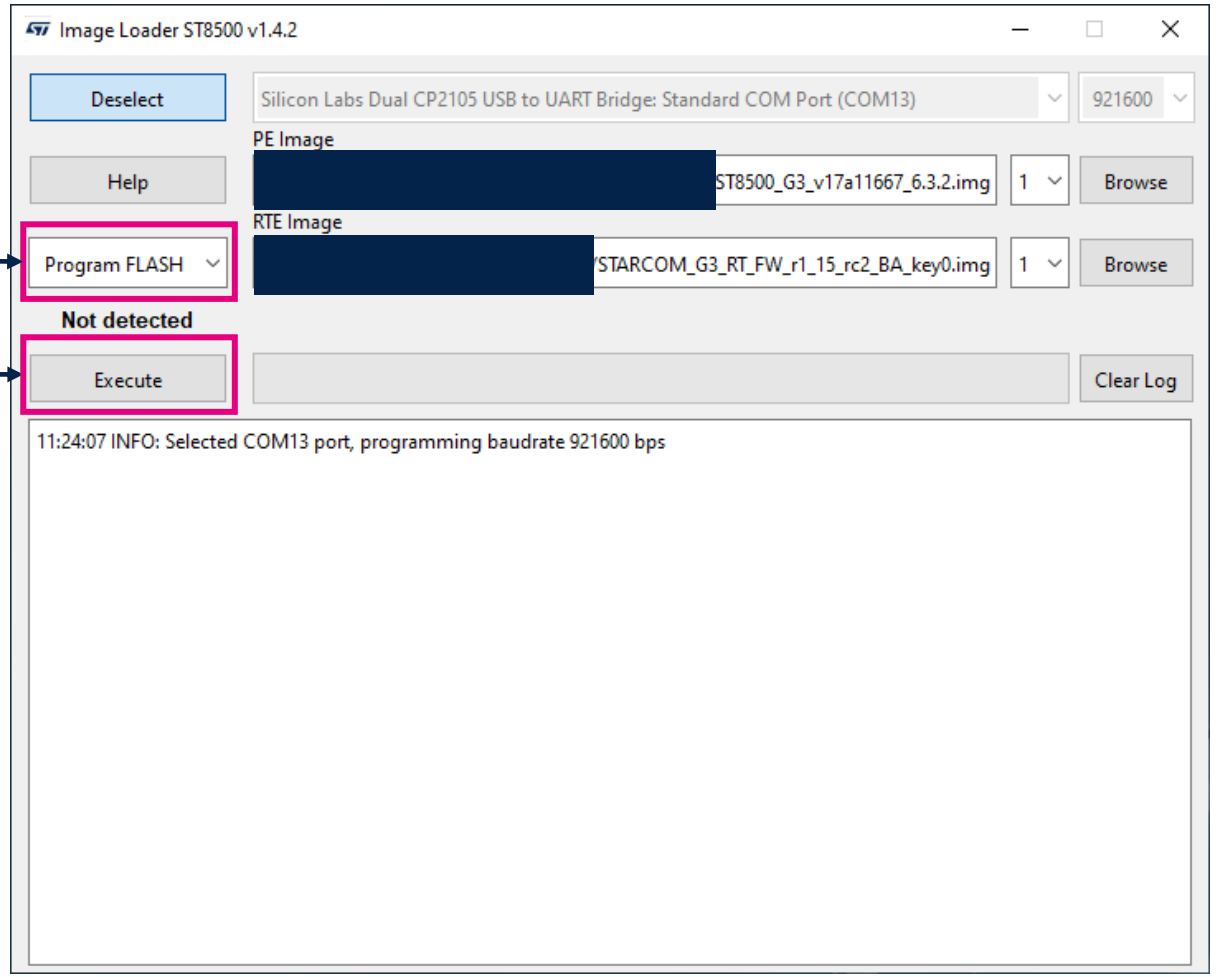
ST8500 FW upgrade execution 2/3

- Program the SPI FLASH memory on the ST8500 evaluation module.

5. Select “Program FLASH”

6. Click on “Execute”

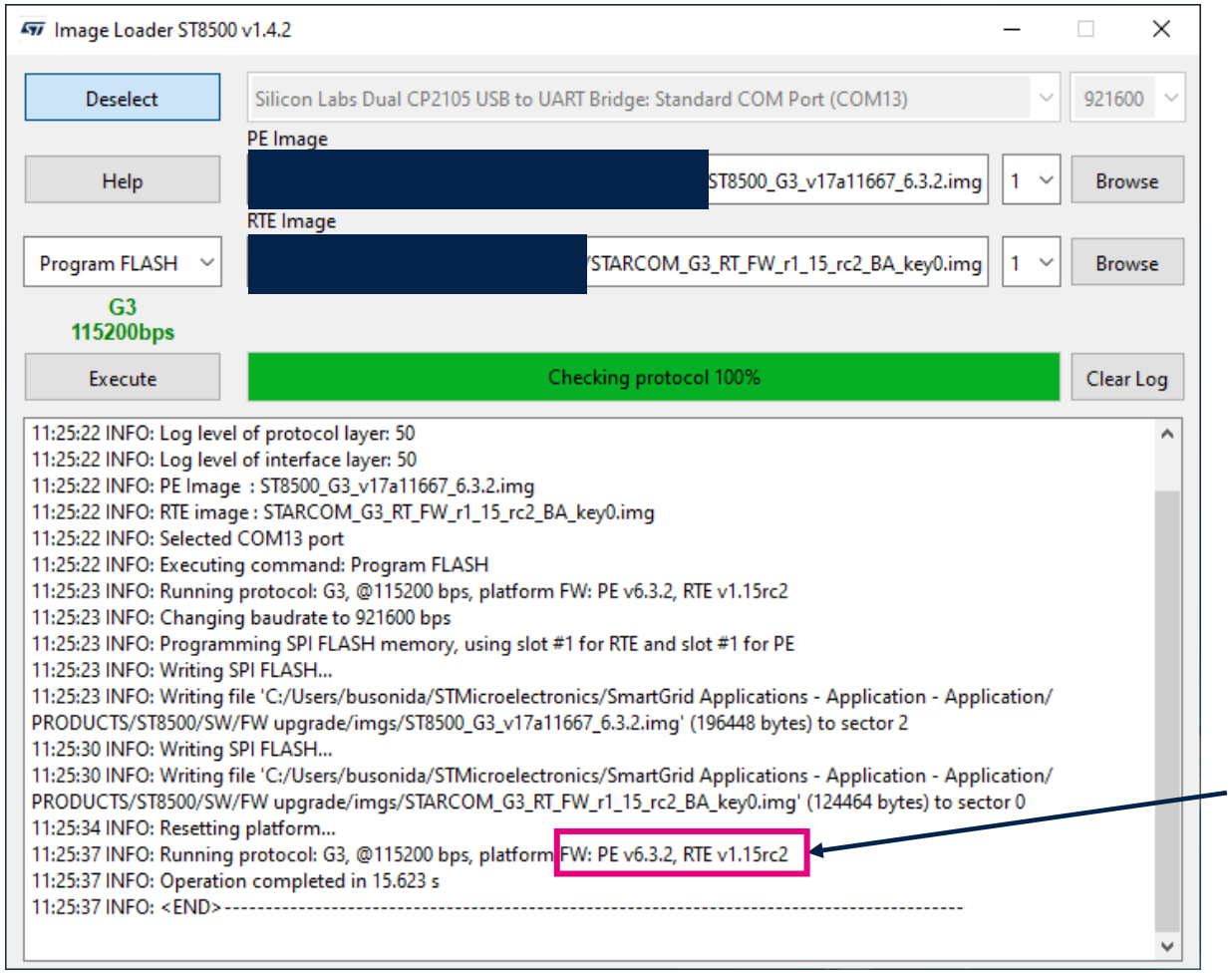
7. Wait for the procedure to complete



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ST8500 FW upgrade execution 3/3

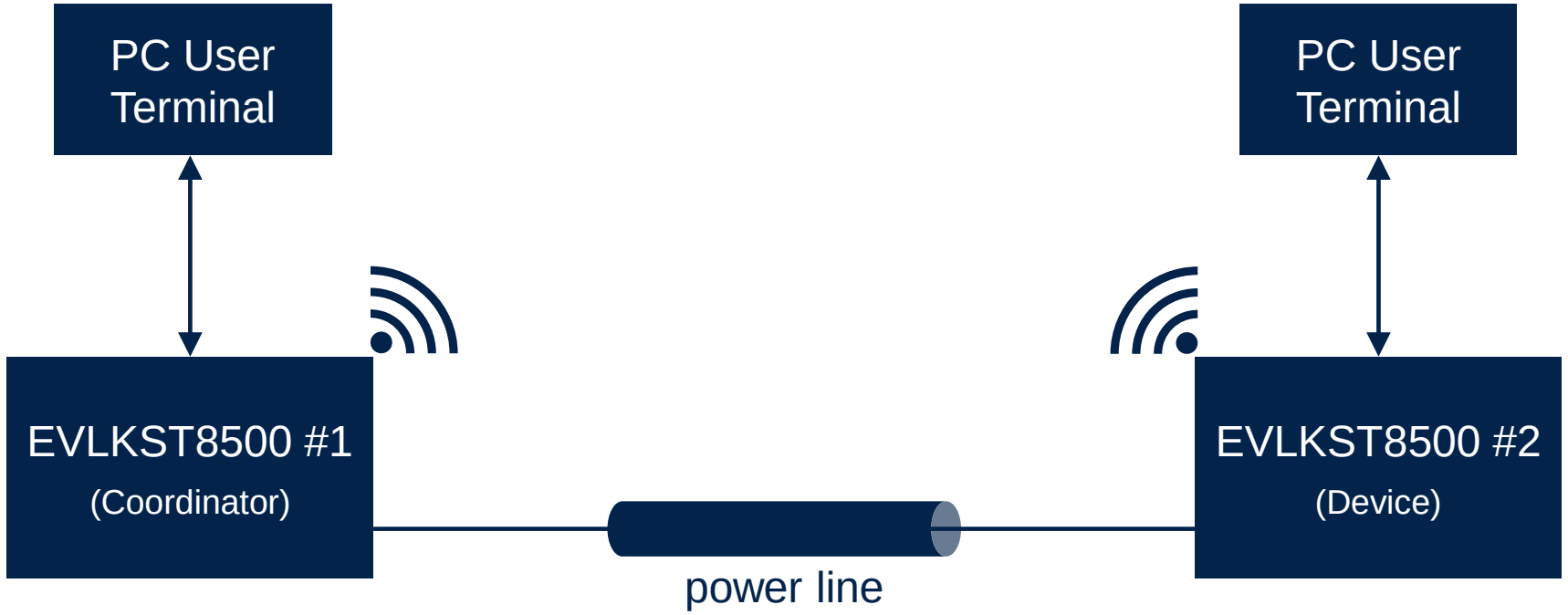
- After the FW upgrade is completed, the Image Loader can be closed.



The FW upgrade is completed, and the programmed FW versions is displayed

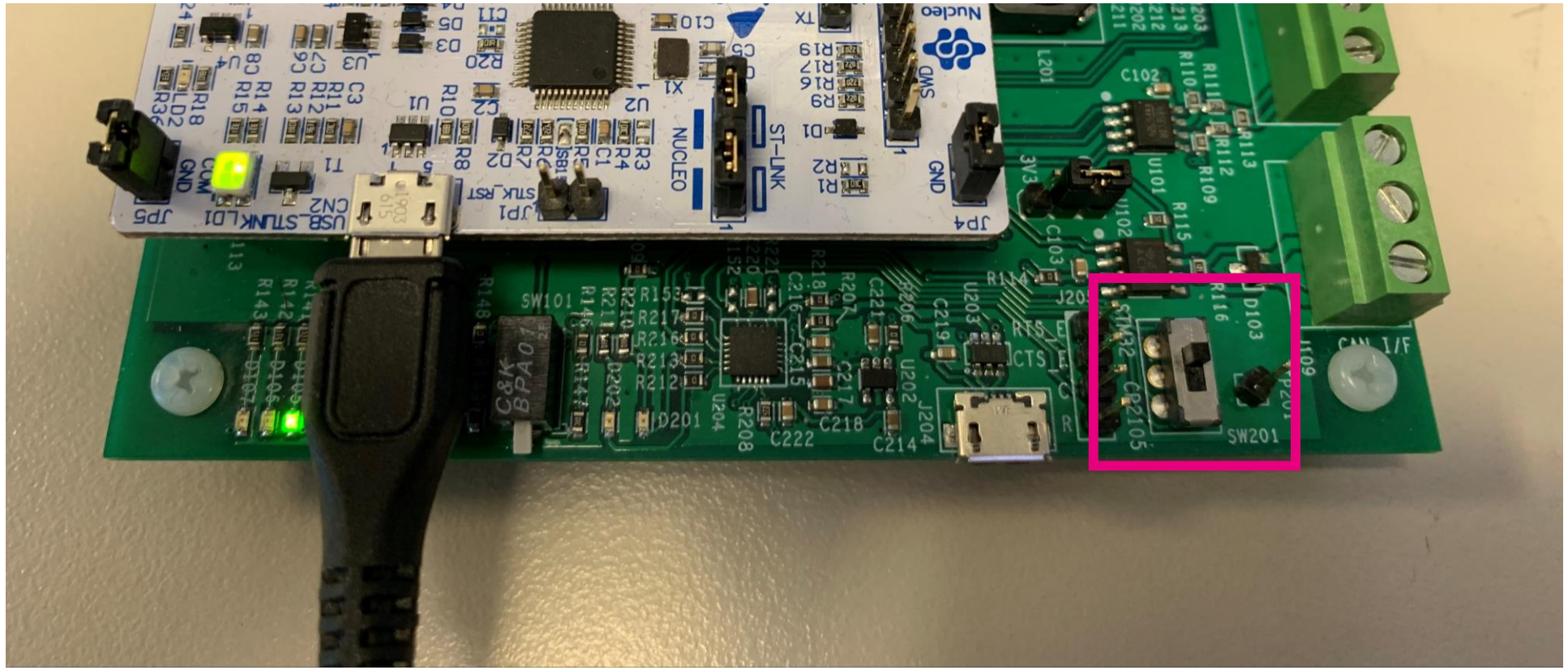
User terminal setup

- Basic/standard setup for the user terminal demo/test comes with 2 G3-Hybrid (PLC/RF) nodes connected as follows:



User terminal preparation

- Connect the PC to the USB socket on the **NUCLEO board**, set SW201 to **STM32**



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General handling

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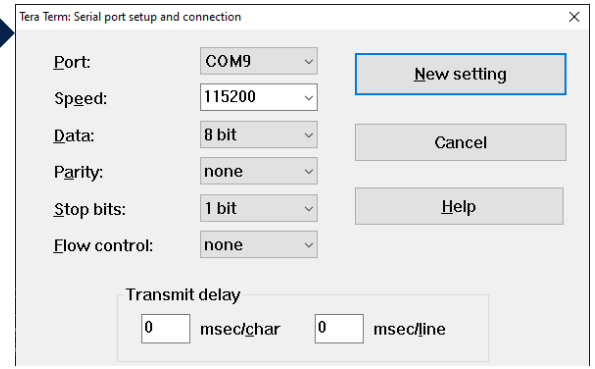
Board configuration

- SW101 switch shall be set DOWN (ST8500 boot from SPI flash) on all boards.
- SW102 switch shall be set DOWN (CENELEC-A mode) on all boards.
- Boards will by default communicate via PLC and/or RF (the medium is chosen depending on the link quality).

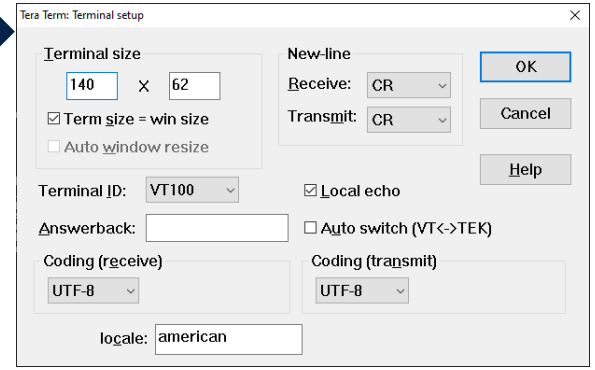
PC Terminal configuration

- Open one serial terminal for each board PC side.
- Serial terminals port configuration:
  - Port: *<COM nb>*
  - Speed: *115200*
  - Data: *8bit*
  - Parity: *none*
  - Stop bits: *1 bit*
  - Flow control: *none*
- Terminal setup:
  - New line: *Receive: CR, Transmit: CR*
  - Coding: *Receive: UTF-8, Transmit: UTF-8*
  - For convenience “Local echo” can be activated (terminal live prints what is typed)

Using Teraterm: menu “Setup -> Serial port…”



Using Teraterm: menu “Setup -> Terminal…”



Terminal startup

- Press the NUCLEO board reset button on all boards: the EVLKST8500 STM32 user terminal prompt appears.
- First, the user is invited to wait (~25-60 s) for the end of the G3-Hybrid Bootstrap phase…

```
0:00:00:00.012576> DEBUG BUILD

0:00:00:00.012661> Kit name: EVLKST8500GH-2
0:00:00:00.012754> STM32G474 FW version: 2.3.0
0:00:00:00.012859> Bandplan: CENELEC A
0:00:00:00.012951> Role: COORDINATOR, RF: 915 MHz
0:00:00:00.013141> STM32 SFlash ID: 0xEF4015

0:00:00:00.013326> Waiting for G3LIB--SW-RESET-------Cnf message...
0:00:00:01.070499> G3LIB--SW-RESET-------Cnf received
0:00:00:01.085092> HOSTIF-TOOLS----------Cnf received

0:00:00:01.085243> Platform info:
0:00:00:01.085331> - Platform: ST8500
0:00:00:01.085423> - FW version: PE v6.3.2, RTE v1.15rc2, G3LIB v17A11667
0:00:00:01.085555> - Role: COORDINATOR
0:00:00:01.085655> - Bandplan: CENELEC A
0:00:00:01.085748> - Mode: IPV6 ADP
0:00:00:01.085840> - RF connected: YES
0:00:00:01.085975> - Platform SFlash ID: 0xEF4015
0:00:00:01.086124> - USART configuration: 01
0:00:00:01.086218> - LED configuration: 01
0:00:00:01.086312> - MAC address: 0080E1FFFE4CA48C

0:00:00:01.257050> HOSTIF-RF-CONFIG-SET--Cnf received
0:00:00:01.257175> G3BOOT-SERVER-STOP----Cnf received
0:00:00:01.257280> Waiting for G3BOOT-SERVER-START---Cnf message...
0:00:00:03.320106> G3BOOT-SERVER-START---Cnf received
0:00:00:03.320215> Waiting for G3BOOT-SERVER-JOIN----Ind message...
```

```
0:00:00:00.014532> DEBUG BUILD

0:00:00:00.014758> Kit name: EVLKST8500GH-2
0:00:00:00.015014> STM32G070 FW version: 2.3.0
0:00:00:00.015297> Bandplan: CENELEC A
0:00:00:00.015542> Role: DEVICE, RF: 915 MHz
0:00:00:00.015935> STM32 SFlash ID: 0xEF4015

0:00:00:00.016406> Waiting for G3LIB--SW-RESET-------Cnf message...
0:00:00:01.075300> G3LIB--SW-RESET-------Cnf received
0:00:00:01.082963> HOSTIF-TOOLS----------Cnf received

0:00:00:01.083356> Platform info:
0:00:00:01.083597> - Platform: ST8500
0:00:00:01.083847> - FW version: PE v6.3.2, RTE v1.15rc2, G3LIB v17A11667
0:00:00:01.084232> - Role: DEVICE
0:00:00:01.084476> - Bandplan: CENELEC A
0:00:00:01.084724> - Mode: IPV6 ADP
0:00:00:01.084973> - RF connected: YES
0:00:00:01.085320> - Platform SFlash ID: 0xEF4015
0:00:00:01.085697> - USART configuration: 01
0:00:00:01.085954> - LED configuration: 01
0:00:00:01.086210> - MAC address: 500F59FFFE31AB29

0:00:00:01.255799> HOSTIF-RF-CONFIG-SET--Cnf received
0:00:00:01.256106> Waiting for G3BOOT-DEVICE-START---Cnf message...
```

Left side terminal:
linked to EVLKST8500 #1 (Coordinator)

Right side terminal:
linked to EVLKST8500 #2 (Device)

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Terminal main menu

- As soon as the final bootstrap messages are received (G3BOOT-SRV-JOIN.Ind on Coordinator side and G3BOOT-DEV-START.Cnf on Device side), the User Terminal main menu becomes available:

```
0:00:00:00.012576> DEBUG BUILD

0:00:00:00.012661> Kit name: EVLKST8500GH-2
0:00:00:00.012754> STM32G474 FW version: 2.3.0
0:00:00:00.012859> Bandplan: CENELEC A
0:00:00:00.012951> Role: COORDINATOR, RF: 915 MHz
0:00:00:00.013141> STM32 SFlash ID: 0xEF4015

0:00:00:00.013326> Waiting for G3LIB--SW-RESET-------Cnf message...
0:00:00:01.068983> G3LIB--SW-RESET-------Cnf received
0:00:00:01.083583> HOSTIF-TOOLS----------Cnf received

0:00:00:01.083734> Platform info:
0:00:00:01.083822> - Platform: ST8500
0:00:00:01.083914> - FW version: PE v6.3.2, RTE v1.15rc2, G3LIB v17A11667
0:00:00:01.084049> - Role: COORDINATOR
0:00:00:01.084151> - Bandplan: CENELEC A
0:00:00:01.084244> - Mode: IPV6 ADP
0:00:00:01.084336> - RF connected: YES
0:00:00:01.084471> - Platform SFlash ID: 0xEF4015
0:00:00:01.084615> - USART configuration: 01
0:00:00:01.084709> - LED configuration: 01
0:00:00:01.084803> - MAC address: 0080E1FFFE4CA48C

0:00:00:01.255549> HOSTIF-RF-CONFIG-SET--Cnf received
0:00:00:01.255671> G3BOOT-SERVER-STOP----Cnf received
0:00:00:01.255777> Waiting for G3BOOT-SERVER-START---Cnf message...
0:00:00:03.323164> G3BOOT-SERVER-START---Cnf received
0:00:00:03.323273> Waiting for G3BOOT-SERVER-JOIN----Ind message...
0:00:00:22.227333> G3BOOT-SERVER-GETPSK--Ind received
0:00:00:22.227454> G3BOOT-SERVER-SETPSK--Cnf received
0:00:00:22.753813> G3BOOT-SERVER-JOIN----Ind received

0:00:00:22.753966> Connection info:
0:00:00:22.754054> - PAN ID: 781D
0:00:00:22.754148> - Short address: 0

0:00:00:22.754302> Waiting for G3UDP--CONNECTION-SET-Cnf message...
0:00:00:22.756196> G3UDP--CONNECTION-SET-Cnf received
0:00:00:22.758025> G3UDP--CONNECTION-SET-Cnf received
0:00:00:22.759798> G3UDP--CONNECTION-SET-Cnf received

0:00:00:22.759963> < Main menu >

0:00:00:22.760051> 1) UDP tests
0:00:00:22.760143> 2) UDP connections
0:00:00:22.760234> 3) Connected devices
0:00:00:22.760325> 4) UDP image transfer
0:00:00:22.760416> 5) Manage SFLASH memory
0:00:00:22.760508> 6) Kick connected device
0:00:00:22.760599> 7) Update GMK
0:00:00:22.760689> 8) Reset

0:00:00:22.760821> Press ESCAPE key then ENTER at any time to go back to this menu
0:00:00:22.760913> Digit the number of an option and then ENTER to select it
```

```
0:00:00:00.014532> DEBUG BUILD

0:00:00:00.014758> Kit name: EVLKST8500GH-2
0:00:00:00.015014> STM32G070 FW version: 2.3.0
0:00:00:00.015297> Bandplan: CENELEC A
0:00:00:00.015542> Role: DEVICE, RF: 915 MHz
0:00:00:00.015935> STM32 SFlash ID: 0xEF4015

0:00:00:00.016406> Waiting for G3LIB--SW-RESET-------Cnf message...
0:00:00:01.075021> G3LIB--SW-RESET-------Cnf received
0:00:00:01.082671> HOSTIF-TOOLS----------Cnf received

0:00:00:01.083068> Platform info:
0:00:00:01.083309> - Platform: ST8500
0:00:00:01.083555> - FW version: PE v6.3.2, RTE v1.15rc2, G3LIB v17A11667
0:00:00:01.083940> - Role: DEVICE
0:00:00:01.084187> - Bandplan: CENELEC A
0:00:00:01.084436> - Mode: IPV6 ADP
0:00:00:01.084681> - RF connected: YES
0:00:00:01.085032> - Platform SFlash ID: 0xEF4015
0:00:00:01.085408> - USART configuration: 01
0:00:00:01.085662> - LED configuration: 01
0:00:00:01.085918> - MAC address: 500F59FFFE31AB29

0:00:00:01.255511> HOSTIF-RF-CONFIG-SET--Cnf received
0:00:00:01.255819> Waiting for G3BOOT-DEVICE-START---Cnf message...
0:00:00:19.330619> G3BOOT-DEVICE-PANSORT-Ind received
0:00:00:19.330950> G3BOOT-DEVICE-PANSORT-Cnf received
0:00:00:24.451932> G3BOOT-DEVICE-START---Cnf received

0:00:00:24.452330> Connection info:
0:00:00:24.452561> - PAN ID: 781D
0:00:00:24.452814> - Short address: 1
0:00:00:24.453066> - LBA address: 0
0:00:00:24.453313> - LBA media type: PLC

0:00:00:24.453692> Waiting for G3UDP--CONNECTION-SET-Cnf message...
0:00:00:24.455922> G3UDP--CONNECTION-SET-Cnf received
0:00:00:24.457736> G3UDP--CONNECTION-SET-Cnf received
0:00:00:24.459478> G3UDP--CONNECTION-SET-Cnf received

0:00:00:24.459896> < Main menu >

0:00:00:24.460129> 1) UDP tests
0:00:00:24.460372> 2) UDP connections
0:00:00:24.460618> 3) Connected devices
0:00:00:24.460869> 4) UDP image transfer
0:00:00:24.461120> 5) Manage SFLASH memory
0:00:00:24.461369> 6) Disconnect/reconnect
0:00:00:24.461619> 7) Enable/disable Fast Restore
0:00:00:24.461875> 8) Reset

0:00:00:24.462215> Press ESCAPE key then ENTER at any time to go back to this menu
0:00:00:24.462473> Digit the number of an option and then ENTER to select it
```

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Terminal test menu

- To select an option of the User Terminal, digit its number and then press ENTER to confirm the selection. Some features require to be selected on more than one board (UDP tests, UDP image transfer…)

```
0:00:00:21.050430> < Main menu >

0:00:00:21.050515> 1) UDP tests
0:00:00:21.050605> 2) UDP connections
0:00:00:21.050696> 3) Connected devices
0:00:00:21.050788> 4) UDP image transfer
0:00:00:21.050879> 5) Manage SFLASH memory
0:00:00:21.050971> 6) Kick connected device
0:00:00:21.051067> 7) Update GMK
0:00:00:21.051159> 8) Reset
```

- At any time, it is possible to abort the current selection and go back to the main menu by pressing ESCAPE and then ENTER.

UDP tests:
single ‘basic’ UDP test

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Single ‘basic’ UDP test launch

- This test aims at making a single UDP data transfer from one node to another.
- The test can be selected from terminals as follows:
  - From *EVLKST8500 main menu*: press **1** to select “*UDP tests”.*
  - Then, from *UDP test menu*: press **1** to select *“Single basic UDP test”.*
- Afterwards, the user is then invited to choose which node will play the role of UDP data originator and which node will play the role of UDP data recipient.
- On originator side, the destination short address must be inserted.

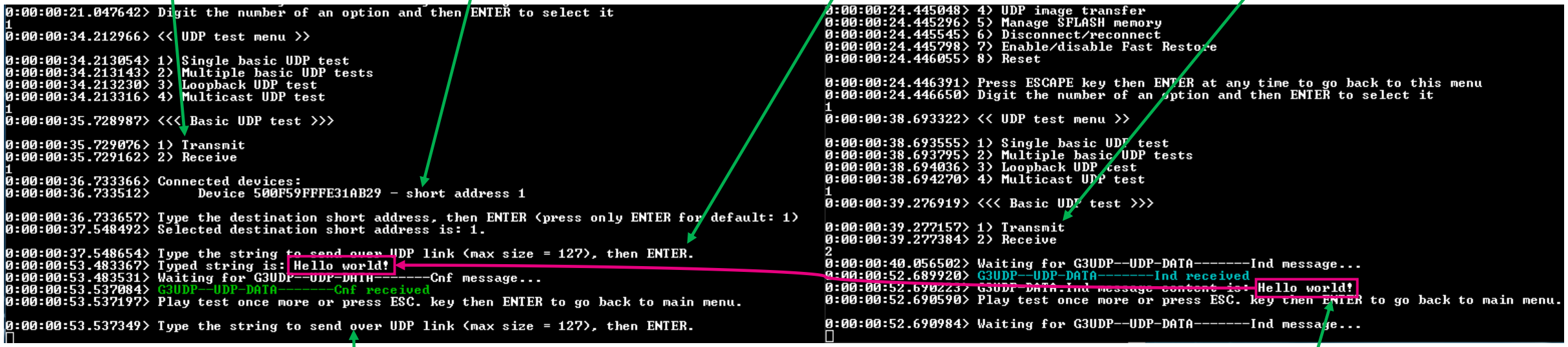
Single ‘basic’ UDP test example

*‘1’ is selected: EVLKST8500 #1 is UDP data **originator***

‘1’ is selected as destination short address (EVLKST8500 #2)

Then user is invited to enter the string to send (string must be ended by pressing ENTER)

*‘2’ is selected: EVLKST8500 #1 is UDP data **recipient***



Test is finished, system can repeat the test of go back to “EVLKST8500 main menu”

Message is received and displayed

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UDP tests:
multiple ‘basic’ UDP test

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Multiple ‘basic’ UDP test launch

- This test aims at making multiple UDP data transfer from one node to another.
- The test can be selected from terminals as follows:
  - From *EVLKST8500 main menu*: press **1** to select “*UDP tests”.*
  - Then, from *UDP test menu*: press **2** to select “*Multiple basic UDP tests”.*
- Afterwards, the user is then invited to choose which node will play the role of UDP data originator and which node will play the role of UDP data recipient.
- On originator side, the destination short address must be inserted.

Multiple ‘basic’ UDP test example

*‘1’ is selected: EVLKST8500 #1 is UDP data **originator***

‘1’ is selected as destination short address (EVLKST8500 #2)

Then user is invited to enter the number of packets (here, 10 is selected) and the string to send (string must be ended by pressing ENTER)

*‘2’ is selected: EVLKST8500 #2 is UDP data **recipient***

EVLKST8500 #2 receives 10 messages…

```
0:00:04:20.939106> << UDP test menu >>

0:00:04:20.939192> 1) Single basic UDP test
0:00:04:20.939278> 2) Multiple basic UDP tests
0:00:04:20.939365> 3) Loopback UDP test
0:00:04:20.939451> 4) Multicast UDP test
2
0:00:04:21.427429> <<< Multiple basic UDP tests >>>

0:00:04:21.427517> 1) Transmit
0:00:04:21.427601> 2) Receive
1
0:00:04:22.508535> Connected devices:
0:00:04:22.508681>       Device 500F59FFFE31AB29 - short address 1

0:00:04:22.508826> Type the destination short address, then ENTER (press only ENTER for default: 1)
0:00:04:23.670742> Selected destination short address is: 1.

0:00:04:23.670904> Enter the number of packets to send (between 1 and 99) then ENTER.
10
0:00:04:34.282764> Type the string to send over UDP link (max size = 127), then ENTER.
0:00:04:42.705405> Typed string is: Sample packet
0:00:04:42.755647> G3UDP--UDP-DATA-------Cnf received
0:00:04:42.823121> G3UDP--UDP-DATA-------Cnf received
0:00:04:42.892214> G3UDP--UDP-DATA-------Cnf received
0:00:04:42.956936> G3UDP--UDP-DATA-------Cnf received
0:00:04:43.023314> G3UDP--UDP-DATA-------Cnf received
0:00:04:43.085871> G3UDP--UDP-DATA-------Cnf received
0:00:04:43.150051> G3UDP--UDP-DATA-------Cnf received
0:00:04:43.220784> G3UDP--UDP-DATA-------Cnf received
0:00:04:43.292611> G3UDP--UDP-DATA-------Cnf received
0:00:04:43.360610> G3UDP--UDP-DATA-------Cnf received

0:00:04:43.360765> Test completed, 10 UDP packets have been sent with the specified data.
0:00:04:43.360865> Play test once more or press ESC. key then ENTER to go back to main menu.

0:00:04:43.361020> Enter the number of packets to send (between 1 and 99) then ENTER.
```

```
0:00:04:18.338729> 4) UDP image transfer
0:00:04:18.338981> 5) Manage SFLASH memory
0:00:04:18.339234> 6) Disconnect/reconnect
0:00:04:18.339484> 7) Enable/disable Fast Restore
0:00:04:18.339738> 8) Reset

0:00:04:18.340081> Press ESCAPE key then ENTER at any time to go back to this menu
0:00:04:18.340340> Digit the number of an option and then ENTER to select it
1
0:00:04:24.820008> << UDP test menu >>

0:00:04:24.820242> 1) Single basic UDP test
0:00:04:24.820479> 2) Multiple basic UDP tests
0:00:04:24.820718> 3) Loopback UDP test
0:00:04:24.820957> 4) Multicast UDP test
2
0:00:04:25.266939> <<< Multiple basic UDP tests >>>

0:00:04:25.267185> 1) Transmit
0:00:04:25.267413> 2) Receive
2
0:00:04:26.282471> Waiting for G3UDP--UDP-DATA-------Ind message...
0:00:04:41.732508> G3UDP--UDP-DATA-------Ind received
0:00:04:41.800462> G3UDP--UDP-DATA-------Ind received
0:00:04:41.869361> G3UDP--UDP-DATA-------Ind received
0:00:04:41.933914> G3UDP--UDP-DATA-------Ind received
0:00:04:42.000631> G3UDP--UDP-DATA-------Ind received
0:00:04:42.062992> G3UDP--UDP-DATA-------Ind received
0:00:04:42.126981> G3UDP--UDP-DATA-------Ind received
0:00:04:42.197515> G3UDP--UDP-DATA-------Ind received
0:00:04:42.267785> G3UDP--UDP-DATA-------Ind received
0:00:04:42.336981> G3UDP--UDP-DATA-------Ind received
0:00:04:42.337296> G3UDP-DATA.Ind message content is: Sample packet
0:00:04:42.337664> Test completed, G3UDP-DATA.Ind received 10 times.
0:00:04:42.337935> Play test once more or press ESC. key then ENTER to go back to main menu.

0:00:04:42.338329> Waiting for G3UDP--UDP-DATA-------Ind message...
```

Then EVLKST8500 #1 transmits 10 messages…

Test is finished, system can repeat the test or go back to “EVLKST8500 main menu”

Number of received messages is displayed, in addition to the last message content

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UDP tests:
‘loopback’ UDP test

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‘Loopback’ UDP test launch

- This test aims at making a UDP data transfer from one node to another and then sending it back to the originator. The data loopback transfer duration and number of exchanged bytes are reported.
- The test can be selected from terminals as follows:
  - From the *main menu*: press **1** to select “*UDP tests”.*
  - Then, from *UDP test menu*: press **3** to select “*Loopback UDP test”.*
- Afterwards, the user is then invited to choose which node will play the role of UDP data originator and which node will play the role of UDP data mirror.
- On originator side, the destination short address must be inserted.

‘Loopback’ UDP test example

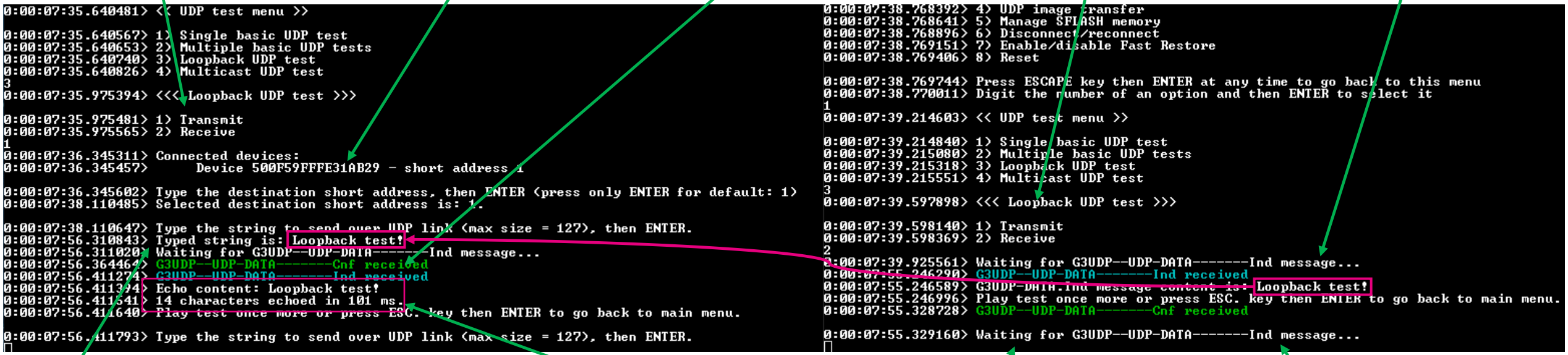
*‘1’ is selected: EVLKST8500 #1 is the UDP data **originator***

‘1’ is selected as destination short address (EVLKST8500 #2)

Then EVLKST8500 #1 sends the typed message

*‘2’ is selected: EVLKST8500 #2 is the UDP data **“mirror”***

Then EVLKST8500 #2 waits for message from the originator…



The user is invited to enter the string to send (string must be ended by pressing ENTER)

Message is received back.
The following info is displayed:
- ***message content***
- ***number of exchanged bytes***
- ***duration of the transfer***

Data from originator is received and sent back

Test is finished, system waits for another packet to retransmit back

life.augmented

UDP tests:
multicast UDP test

life.augmented

Multicast UDP test launch

- This test aims at making a UDP data transfer in broadcast. The data is sent to all nearby nodes by using a special IPv6 destination address.
- The test can be selected from terminals as follows:
  - From the *main menu*: press **1** to select "*UDP tests".*
  - Then, from *UDP test menu*: press **4** to select "*Multicast UDP test".*
- Afterwards, the user is then invited to choose which node will play the role of UDP data transmitter and which node will play the role of UDP data receiver.
- On originator side, there is no need to specify the destination short address since the UDP packet is sent in broadcast.

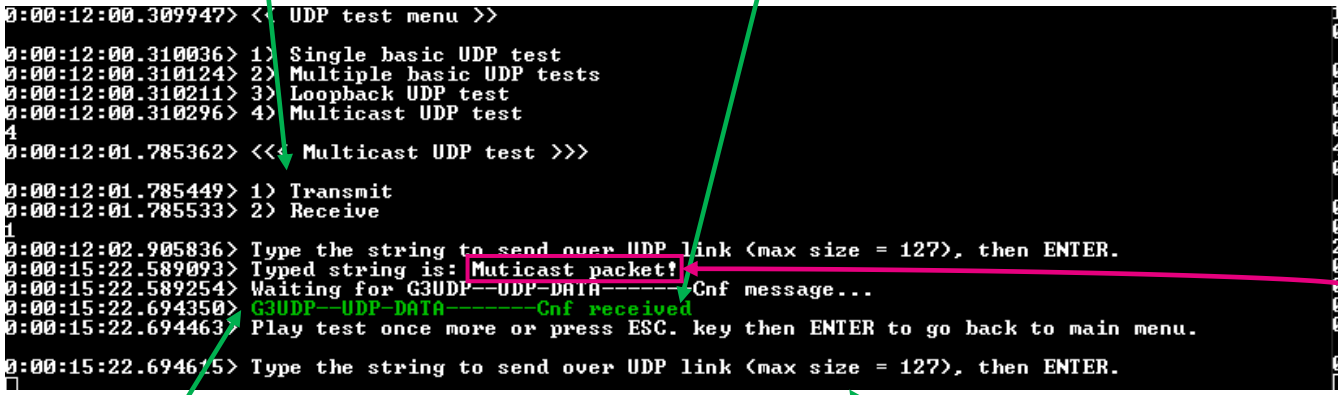
Multicast UDP test example

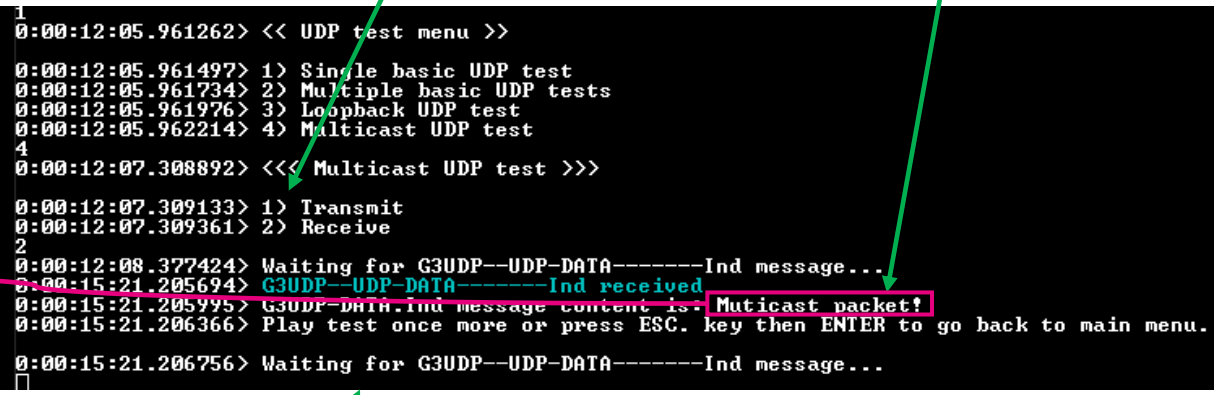
*'1' is selected: EVLKST8500 #1 is the UDP data **transmitter***

Then EVLKST8500 #1 sends the typed message

*'2' is selected: EVLKST8500 #2 is the UDP data **receiver***

Then EVLKST8500 #2 waits for message from the originator…





The user is invited to enter the string to send (string must be ended by pressing ENTER)

Test is finished, system can repeat the test or go back to "EVLKST8500 main menu"

Data from originator is received

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UDP connections

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UDP connections launch

- This feature can be used to display the platform information and all current UDP connections settings.
- The feature can be selected from terminals as follows:
  - From the *main menu*: press **2** to select "*UDP connections".*
- Afterwards, the platform information and all configured connections are printed on the terminal.

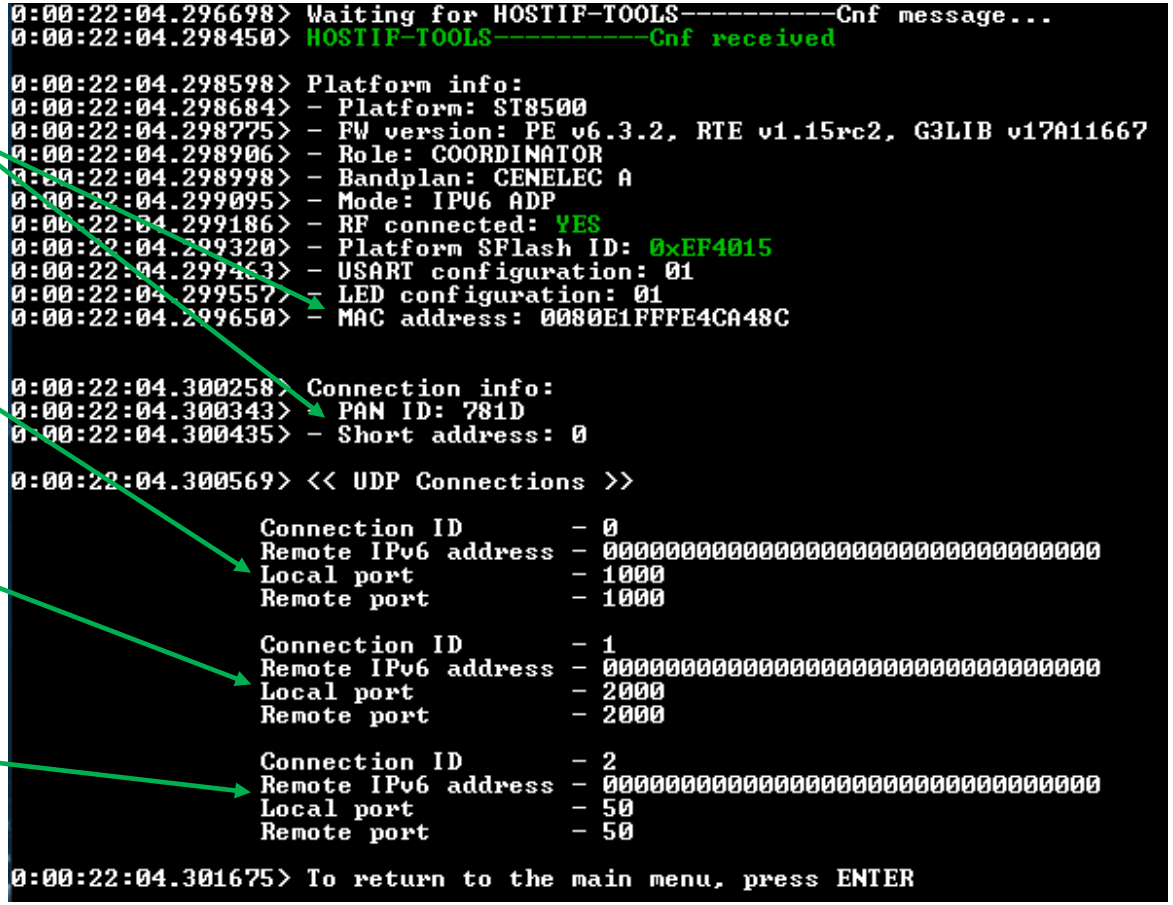
UDP connections example

These are the MAC address, the PAN ID and the short address of this device

These are the ID, the IPv6 address, the local port and the remote port of the test connection, used for UDP tests.

These are the ID, the IPv6 address, the local port and the remote port of the transfer connection, used by the UDP image transfer module

These are the ID, the IPv6 address, the local port and the remote port of the Last Gasp connection, used by the G3 Last Gasp module



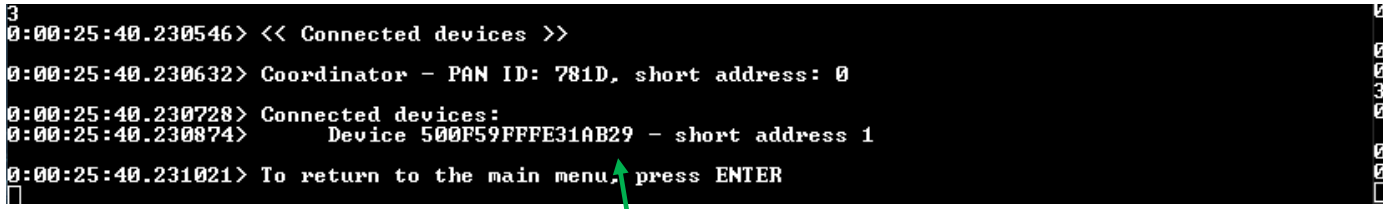
Connected devices

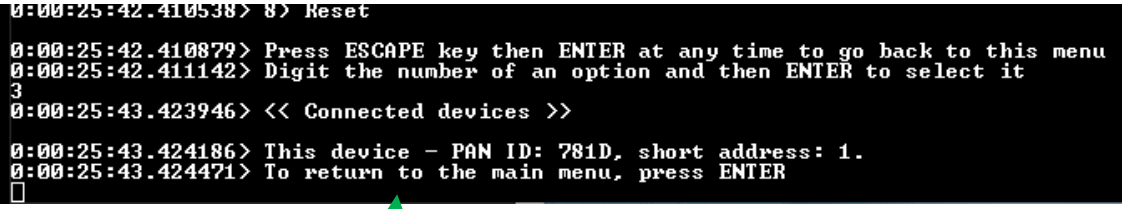
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Connected devices launch

- This feature can be used to display all current connected devices.
- The feature can be selected from terminals as follows:
  - From the *main menu*: press **3** to select "*Connected devices".*
- Afterwards, all connected devices are printed on the terminal.
- PAN coordinators display the devices connected to their PAN.
- Devices display their PAN ID and short address.

Connected devices example





'EVLKST8500 #1 is a PAN coordinator EVLKST8500 #2 is connected to its PAN with short address 1

'EVLKST8500 #2 is a PAN device. It is connected to the PAN with short address 1

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UDP image transfer

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UDP image transfer launch

- This optional feature starts the procedure to transfer a ST8500 image (PE or RTE) from the coordinator to a connected device.
  - To use this feature, “ENABLE_IMAGE_TRANSFER” macro in *settings.h* must be set to 1.
- The feature can be selected from terminals as follows:
  - From the m*ain menu*: press **4** to select *the “UDP image transfer”.*
- Afterwards, user can follow the steps prompted by the terminal.
- The option needs to be selected on both coordinator and device.
- The coordinator must have a ST8500 image stored in the SPI flash memory (it can be stored using the External Loader in STM32CubeProgrammer).

UDP image transfer example

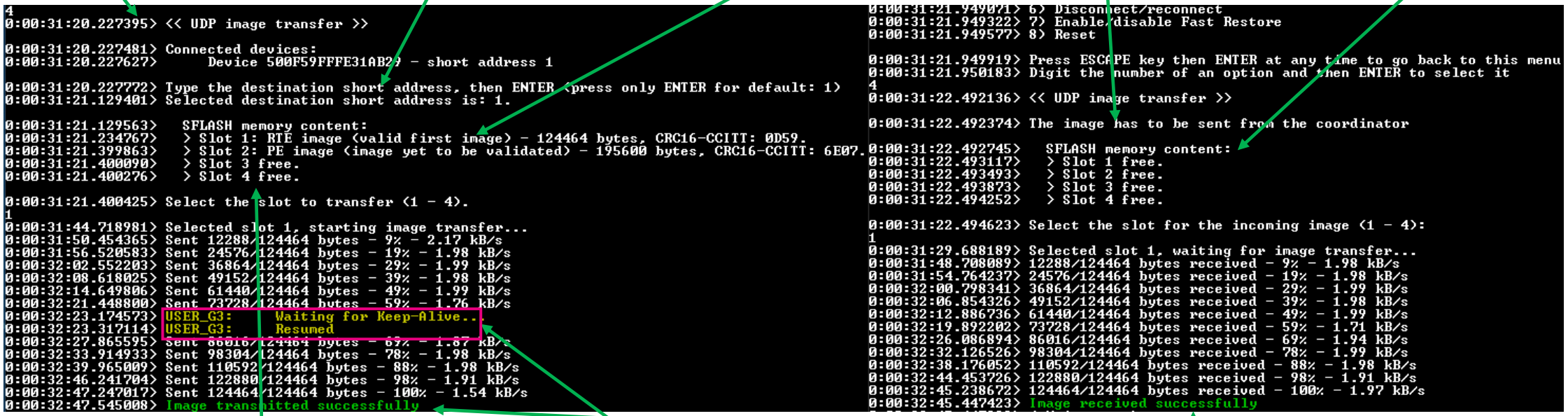
*EVLKST8500 #1 is the UDP image **sender (coordinator)***

'1' is selected as destination short address (EVLKST8500 #2)

'1' is selected: the RTE image in the first slot shall be sent

*EVLKST8500 #2 is the UDP image **receiver** (device)*

'1' is selected: the incoming image shall be stored in first slot



The SPI flash memory content is displayed: it is divided in 4 slots of 256 KB, each slot can contain an image

Note: only half of the memory is used to store images

***Transfer is completed**: the CRC16 check of the received image was successful*

The UDP image transfer shares the use of the IPv6 layer with the G3 Keep-Alive module, which has higher priority. This warning messages are harmless and completely normal

life.augmented

Manage SPI Flash memory

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Manage SPI Flash memory launch

- This optional feature can be used to clean the SPI Flash memory of the STM32.
  - To use this feature, “ENABLE_SFLASH_MANAGEMENT” macro in *settings.h* must be set to 1.
- The feature can be selected from terminals as follows:
  - From the *main menu*: press **5** to select “*Manage SFLASH memory”.*
- Afterwards, the user can choose to erase a single slot or the entire memory.

Manage SPI Flash memory example

The SPI flash memory content is displayed: it is divided in 4 slots of 256 KB, each slot can contain an image

Note: only half of the memory is used to store images

‘1’ is selected: the first slot is then erased

Erasure is completed after few seconds

```
5
0:00:29:31.309198> << SFLASH management >>
0:00:29:31.309529>   SFLASH memory content:
0:00:29:31.510554>    > Slot 1: PE image (valid second image) - 195600 bytes, CRC16-CCITT: 6E07.
0:00:29:31.511051>    > Slot 2 free.
0:00:29:31.511427>    > Slot 3 free.
0:00:29:31.511806>    > Slot 4 free.

0:00:29:31.512154> Press 'y' then ENTER to erase all slots (1 - 4)
0:00:29:31.512404> Press '1', '2', '3', '4', then ENTER to erase a single slot
0:00:29:31.512662> Press 'm' then ENTER to erase the NVM
0:00:29:31.512910> Press 'n' then ENTER to abort
0:00:29:49.492556> Erasing slot 1...
0:00:29:49.506854> Erasure complete
```

Kick/Connect/Disconnect

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Kick/Connect/Disconnect launch

- This optional feature can be used to kick a device out of the PAN (coordinator side) or disconnect/reconnect a device (device side).
  - To use this feature, “ENABLE_DEVICE_MANAGEMENT” macro in *settings.h* must be set to 1.
- The feature can be selected from terminals as follows:
  - From the *coordinator main menu*: press **6** to select “*Kick connected device”.*
  - From the *device main menu*: press **6** to select “*Disconnect/reconnect”.*
- On coordinator side, a kick request will be sent to the selected device.
- On device side, depending on the current connection state, a start request (when disconnected) or leave request (when connected) will be sent.

Kick example

*EVLKST8500 #1 is the **coordinator***

'1' is selected as short address of the device to kick (EVLKST8500 #2)

*EVLKST8500 #2 is the **device***



The confirmation of the kick request is received (and successful) on coordinator side

A device leave indication is received on device side

Disconnect example

*EVLKST8500 #1 is the **coordinator***

*EVLKST8500 #2 is the **device***

'y' is selected to confirm the action



A server leave indication is received on coordinator side

The confirmation of the leave request is received (and successful) on device side

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Connect example

*EVLKST8500 #1 is the **coordinator***

*EVLKST8500 #2 is the **device***

'y' is selected to confirm the action

The device waits for the bootstrap to be completed



A server join indication is received on coordinator side, after the server GETPSK and SETPSK messages (related to the bootstrap)

The confirmation of the device start is received (with success) on device side

GMK update

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GMK update launch

- This optional feature can be used to update the Group Master Key on every device connected to the PAN (and on the PAN coordinator itself).
  - To use this feature, both “ENABLE_BOOT_SERVER_ON_HOST” and “ENABLE_REKEYING” macros in *settings.h* must be set to 1.
- The feature can be selected from coordinator terminal as follows:
  - From *EVLKST8500 coord. main menu*: press **7** to select “*Update GMK”.*
- The GMK is changed on every device, without the need to restart the network.
- The new GMK must be coded manually, in hexadecimal format (such as “11223344556677881122334455667788”).

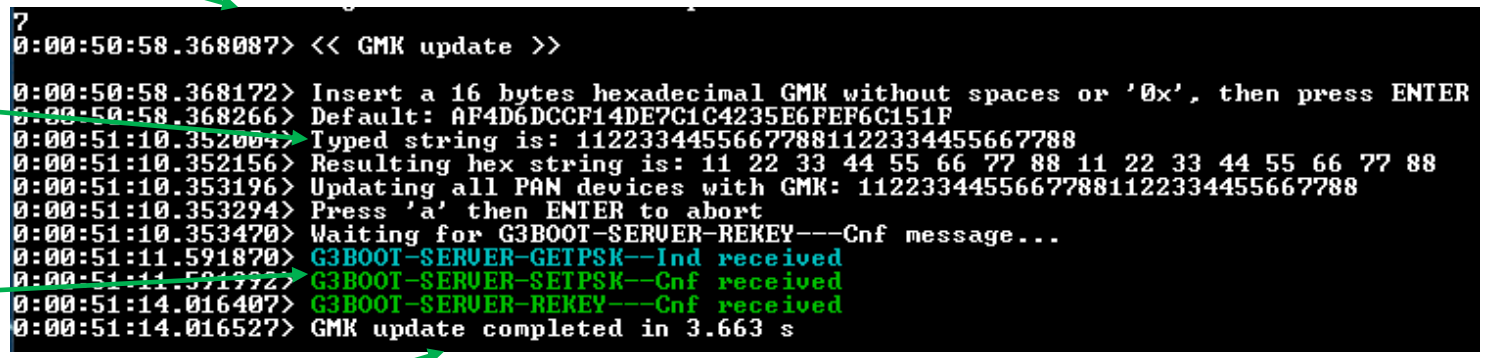
GMK update example

*EVLKST8500 is the **coordinator***

*The new GMK is inserted (16 consecutive hexadecimal values), e.g.: **11223344556677881122334455667788***

The GMK update procedure (described in detail in the AN5715) is executed for all connected devices

*Afterward, all messages exchanged on the PAN are cyphered with the new GMK (**11223344556677881122334455667788**)*



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Fast Restore

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Fast restore enabling/disabling

- This optional feature can be used to enable or disable the Fast Restore mode on a PAN Device.
  - To use this feature, the “ENABLE_FAST_RESTORE” macro in *settings.h* must be set to 1.
- The feature can be selected from device terminal as follows:
  - From *EVLKST8500 device main menu*: press **7** to select “*Enable/disable Fast Restore”.*
- This setting is stored inside the NVM, and it is applied at the next reboot.
- The Fast Restore mode is used to quickly recover the last connection in case of reboot, without the bootstrap procedure. A functioning connection to a PAN must be established before the Fast Restore is performed.

Fast restore enabling example

*EVLKST8500 is the **device***

The Fast Restore is enabled (inserting 'y' is required to confirm the action)

After resetting the board, the Fast Restore is performed and the connection is restored without the need to repeat the bootstrap

```
7
0:01:02:55.634999> << Fast Restore >>
0:01:02:55.635237> Enable Fast Restore? (y/n)
0:01:02:57.200110> Switch to fast restore boot
```

```
0:00:00:00.014532> DEBUG BUILD

0:00:00:00.014758> Kit name: EVLKST8500GH-2
0:00:00:00.015014> STM32G070 FW version: 2.3.0
0:00:00:00.015297> Bandplan: CENELEC A
0:00:00:00.015542> Role: DEVICE, RF: 915 MHz
0:00:00:00.015935> STM32 SFlash ID: 0xEF4015

0:00:00:00.016406> Waiting for G3LIB--SW-RESET-------Cnf message...
0:00:00:01.077175> G3LIB--SW-RESET-------Cnf received
0:00:00:01.084829> HOSTIF-TOOLS----------Cnf received

0:00:00:01.085222> Platform info:
0:00:00:01.085464> - Platform: ST8500
0:00:00:01.085710> - FW version: PE v6.3.2, RTE v1.15rc2, G3LIB v17A11667
0:00:00:01.086098> - Role: DEVICE
0:00:00:01.086342> - Bandplan: CENELEC A
0:00:00:01.086591> - Mode: IPV6 ADP
0:00:00:01.086839> - RF connected: YES
0:00:00:01.087187> - Platform SFlash ID: 0xEF4015
0:00:00:01.087563> - USART configuration: 01
0:00:00:01.087820> - LED configuration: 01
0:00:00:01.088076> - MAC address: 500F59FFFE31AB29

0:00:00:01.257231> HOSTIF-RF-CONFIG-SET--Cnf received
0:00:00:01.257535> Waiting for G3BOOT-DEVICE-START---Cnf message...
0:00:00:05.537911> G3BOOT-DEVICE-START---Cnf received

0:00:00:05.538309> Connection info:
0:00:00:05.538540> - PAN ID: 781D
0:00:00:05.538794> - Short address: 1
0:00:00:05.539045> - LBA address: 0
0:00:00:05.539292> - LBA media type: PLC

0:00:00:05.539671> Waiting for G3UDP--CONNECTION-SET-Cnf message...
0:00:00:05.541905> G3UDP--CONNECTION-SET-Cnf received
0:00:00:05.543722> G3UDP--CONNECTION-SET-Cnf received
0:00:00:05.545463> G3UDP--CONNECTION-SET-Cnf received
```

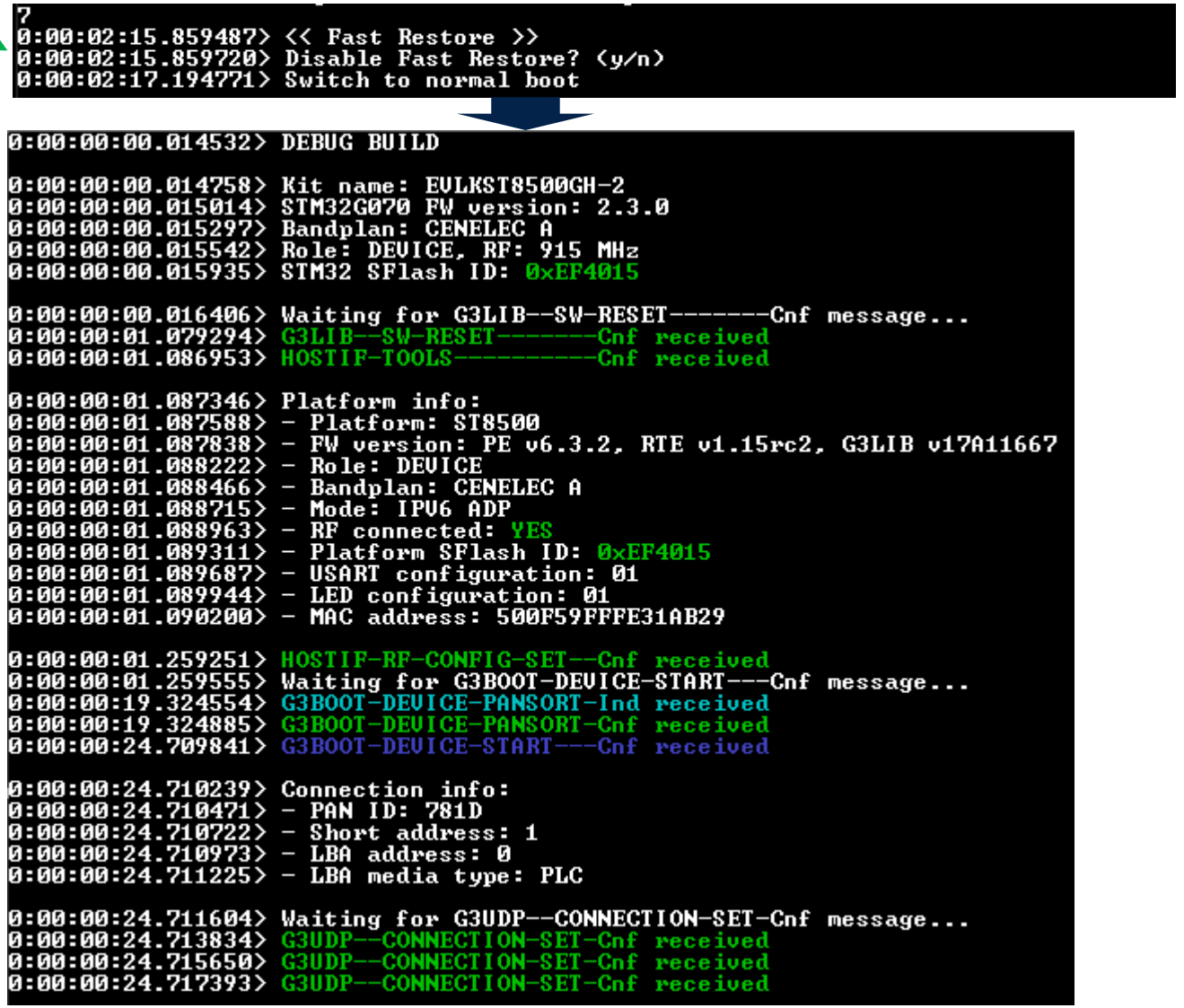
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Fast restore disabling example

*EVLKST8500 is the **device***

The Fast Restore is disabled (inserting 'y' is required to confirm the action)

After resetting the board, the normal boot is performed, and the connection is established with the bootstrap



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Reset

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Reset launch/example

- This feature can be used to reset the STM32 and the ST8500 via software.
- The feature can be selected from terminals as follows:
  - From *EVLKST8500 main menu*: press **8** to select "*Reset".*

```
8
0:00:05:49.876668> << System reset >>

0:00:05:49.876906> Reset the system? (y/n)
0:00:00:00.010087>
0:00:00:00.014532> DEBUG BUILD

0:00:00:00.014758> Kit name: EVLKST8500GH-2
0:00:00:00.015014> STM32G070 FW version: 2.3.0
0:00:00:00.015297> Bandplan: CENELEC A
0:00:00:00.015542> Role: DEVICE, RF: 915 MHz
0:00:00:00.015935> STM32 SFlash ID: 0xEF4015

0:00:00:00.016406> Waiting for G3LIB--SW-RESET-------Cnf message...
0:00:00:01.079796> G3LIB--SW-RESET-------Cnf received
0:00:00:01.087449> HOSTIF-TOOLS----------Cnf received

0:00:00:01.087842> Platform info:
0:00:00:01.088087> - Platform: ST8500
0:00:00:01.088334> - FW version: PE v6.3.2, RTE v1.15rc2, G3LIB v17A11667
0:00:00:01.088715> - Role: DEVICE
0:00:00:01.088962> - Bandplan: CENELEC A
0:00:00:01.089214> - Mode: IPV6 ADP
0:00:00:01.089459> - RF connected: YES
0:00:00:01.089807> - Platform SFlash ID: 0xEF4015
0:00:00:01.090187> - USART configuration: 01
0:00:00:01.090440> - LED configuration: 01
0:00:00:01.090693> - MAC address: 500F59FFFE31AB29

0:00:00:01.259825> HOSTIF-RF-CONFIG-SET--Cnf received
0:00:00:01.260132> Waiting for G3BOOT-DEVICE-START---Cnf message...
```

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Last Gasp

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Last Gasp Launch

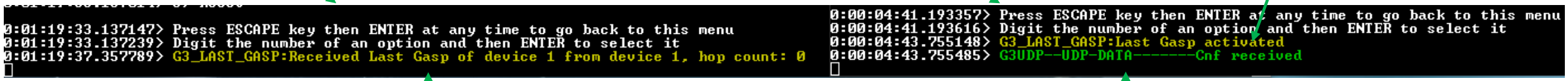
- This optional feature can be used to activate the Last Gasp mode on a PAN Device.
  - To use this feature, the “ENABLE_LAST_GASP” macro in *settings.h* must be set to 1.
- The feature can be selected from device terminal as follows:
  - On the *EVLKST8500 device* kit, press the push-button on the Nucleo board.
- This feature is only a demonstration of the feature. In a real example the Last Gasp should be activated upon detecting a power outage.
- The Last Gasp can be activated only after the device has connected to a PAN.
- Activating the Last Gasp disables the PLC communication, and it can compromise further communication, including the Keep-Alive (it is supposed to be activated shortly before the device switches off)

Last Gasp example

*EVLKST8500 #1 is the **coordinator***

*EVLKST8500 #2 is the **device***

Push-button pressed at 4:43



*Last Gasp message received from the **coordinator***

*Last Gasp message sent from the **device***

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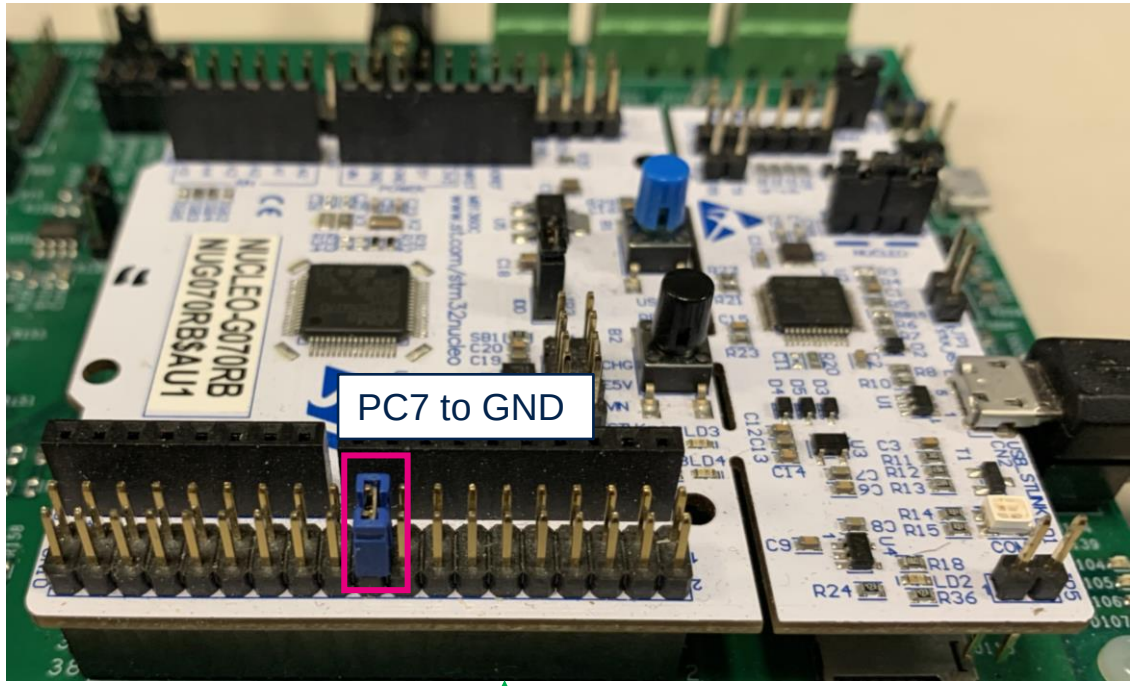
MAC test mode

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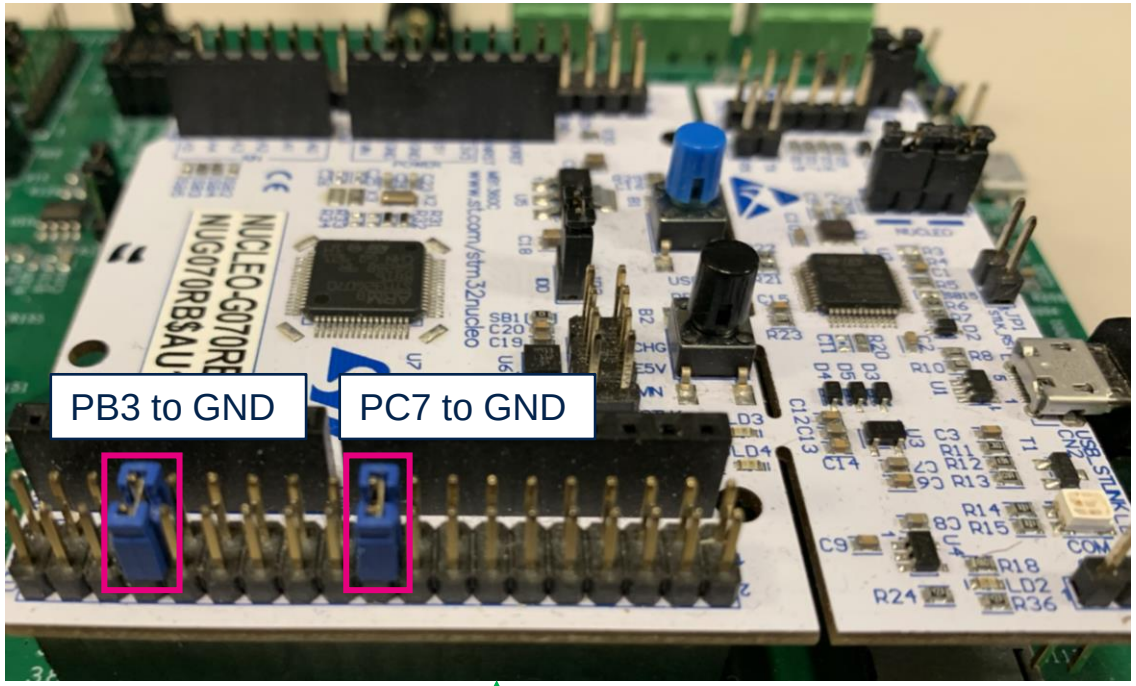
MAC test mode

- This mode is activated at start-up in case a particular GPIO configuration is detected.
- On device side, this mode include various tests on the board HW (Modbus, FAT FS, SPI Flash) that can be disabled or enabled in *settings.h*
- *The MAC test consists in exchanging MAC PLC and RF frames between two boards in MAC mode*
- *The test can be activated with two different methods:*
  - *press* the push-button on the Nucleo board (generic test, for two device only).
  - *insert a command string on the terminal and then ENTER (specific test).*

MAC test mode - GPIO configuration

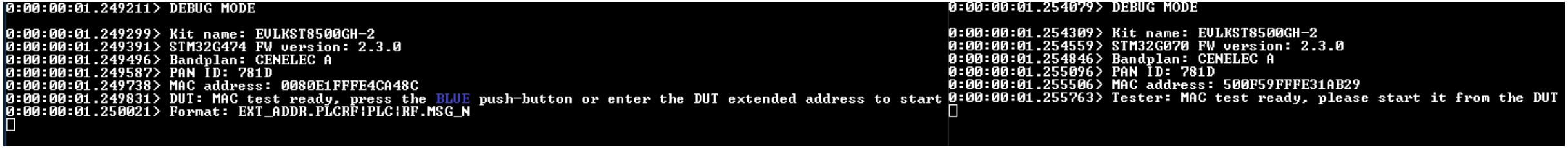


MAC mode enabled, DUT mode



MAC mode enabled, Tester mode

MAC test mode - menu



EVLKST8500 #1 - DUT

EVLKST8500 #2 - Tester

MAC test mode – Launch by push-button 1/2

- First method: test started by pressing the blue push-button on the DUT

```
0:00:00:01.249211> DEBUG MODE

0:00:00:01.249299> Kit name: EVLKST8500GH-2
0:00:00:01.249391> STM32G474 FW version: 2.3.0
0:00:00:01.249496> Bandplan: CENELEC A
0:00:00:01.249587> PAN ID: 781D
0:00:00:01.249738> MAC address: 0080E1FFFE4CA48C
0:00:00:01.249831> DUT: MAC test ready, press the BLUE push-button or enter the DUT extended address to start
0:00:00:01.250021> Format: EXT_ADDR.PLCRF|PLC|RF.MSG_N
0:00:19:29.546150> Starting MAC test...
0:00:19:29.776810> Received 1/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.84 dBuV)
0:00:19:30.030569> Received 2/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.81 dBuV)
0:00:19:30.285354> Received 3/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.84 dBuV)
0:00:19:30.546564> Received 4/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.84 dBuV)
0:00:19:30.809507> Received 5/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.54 dBuV, noise = 75.84 dBuV)
0:00:19:31.068569> Received 6/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.75 dBuV)
0:00:19:31.330664> Received 7/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.82 dBuV)
0:00:19:31.589724> Received 8/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.75 dBuV)
0:00:19:31.846860> Received 9/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.81 dBuV)
0:00:19:32.101014> Received 10/20 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.75 dBuV)
0:00:19:32.259346> Received 11/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:32.413265> Received 12/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:32.568162> Received 13/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:32.723066> Received 14/20 MCPS_DATA_IND (RF): LQI = 161 (RSSI = -13 dBm)
0:00:19:32.883378> Received 15/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:33.037460> Received 16/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:33.196790> Received 17/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:33.350150> Received 18/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:33.509449> Received 19/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:33.665299> Received 20/20 MCPS_DATA_IND (RF): LQI = 160 (RSSI = -14 dBm)
0:00:19:33.665433> Test result: 3
0:00:19:33.665524> MAC test ok
```

EVLKST8500 #1 - DUT

MAC test mode – Launch by push-button 2/2

```
0:00:00:01.254079> DEBUG MODE

0:00:00:01.254309> Kit name: EVLKST8500GH-2
0:00:00:01.254559> STM32G070 FW version: 2.3.0
0:00:00:01.254846> Bandplan: CENELEC A
0:00:00:01.255096> PAN ID: 781D
0:00:00:01.255506> MAC address: 500F59FFFE31AB29
0:00:00:01.255763> Tester: MAC test ready, please start it from the DUT
0:00:07:46.341504> Received 1/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.74 dBuV)
0:00:07:46.594547> Received 2/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.69 dBuV)
0:00:07:46.849507> Received 3/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.88 dBuV)
0:00:07:47.110850> Received 4/20 MCPS_DATA_IND (PLC): LQI = 229 (SNR = 47.25 dB, signal = 122.56 dBuV, noise = 75.51 dBuV)
0:00:07:47.373693> Received 5/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.72 dBuV)
0:00:07:47.632608> Received 6/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.63 dBuV)
0:00:07:47.892508> Received 7/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.63 dBuV)
0:00:07:48.150405> Received 8/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.70 dBuV)
0:00:07:48.410245> Received 9/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.76 dBuV)
0:00:07:48.663467> Received 10/20 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.72 dBuV)
0:00:07:48.843676> Received 11/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:49.000706> Received 12/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:49.156244> Received 13/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:49.310799> Received 14/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:49.467087> Received 15/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:49.623761> Received 16/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:49.781903> Received 17/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:49.936144> Received 18/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:50.092876> Received 19/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
0:00:07:50.252063> Received 20/20 MCPS_DATA_IND (RF): LQI = 170 (RSSI = -4 dBm)
```

EVLKST8500 #2 - Tester

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MAC test mode – Launch by command string 1/2

- Second method: test started by sending a command string through the terminal, on the DUT
- Format: “EXTENDED_ADDRESS.TEST_TYPE.FRAME_NUMBER”
  - TEST_TYPE values: “PLCRF”, “PLC”, “RF”.
  - The string “” is used in the example is “500F59FFFE31AB29.PLC.5”

```
0:00:00:01.248123> DEBUG MODE

0:00:00:01.248210> Kit name: EVLKST8500GH-2
0:00:00:01.248303> STM32G474 FW version: 2.3.0
0:00:00:01.248407> Bandplan: CENELEC A
0:00:00:01.248499> PAN ID: 781D
0:00:00:01.248649> MAC address: 0080E1FFFE4CA48C
0:00:00:01.248743> DUT: MAC test ready, press the BLUE push-button or enter the DUT extended address to start
0:00:00:01.248930> Format: EXT_ADDR.PLCRF|PLC|RF.MSG_N
0:00:01:52.899211> Typed string is: 500F59FFFE31AB29.PLC.5
0:00:01:52.899360> Resulting hex string is: 50 0F 59 FF FE 31 AB 29
0:00:01:52.899957> Testing 5 messages, type: 1, destination: 500F59FFFE31AB29
0:00:01:52.900122> Starting MAC test...
0:00:01:53.134006> Received 1/5 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.86 dBuV)
0:00:01:53.394119> Received 2/5 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.81 dBuV)
0:00:01:53.650725> Received 3/5 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.87 dBuV)
0:00:01:53.907806> Received 4/5 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.84 dBuV)
0:00:01:54.160925> Received 5/5 MCPS_DATA_IND (PLC): LQI = 227 (SNR = 46.75 dB, signal = 122.56 dBuV, noise = 75.81 dBuV)
0:00:01:54.161186> Test result: 3
0:00:01:54.161277> MAC test ok
```

EVLKST8500 #1 - DUT

MAC test mode – Launch by command string 2/2

```
0:00:00:01.254068> DEBUG MODE

0:00:00:01.254298> Kit name: EVLKST8500GH-2
0:00:00:01.254548> STM32G070 FW version: 2.3.0
0:00:00:01.254835> Bandplan: CENELEC A
0:00:00:01.255085> PAN ID: 781D
0:00:00:01.255495> MAC address: 500F59FFFE31AB29
0:00:00:01.255752> Tester: MAC test ready, please start it from the DUT
0:00:01:47.909638> Received 1/5 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.73 dBuV)
0:00:01:48.172913> Received 2/5 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.80 dBuV)
0:00:01:48.429011> Received 3/5 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.54 dBuV, noise = 75.60 dBuV)
0:00:01:48.686026> Received 4/5 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.74 dBuV)
0:00:01:48.939156> Received 5/5 MCPS_DATA_IND (PLC): LQI = 228 (SNR = 47.00 dB, signal = 122.56 dBuV, noise = 75.67 dBuV)
```

EVLKST8500 #2 - Tester

Thank you

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